

**Time:** 5 minutes**Outcome:** Explain how pairwise majority rule can generate cyclic collective preferences.**Difficulty:** ●●○ Intermediate

Condorcet Paradox & Voting Cycles



THE CORE IDEA

Even when every voter has a consistent ranking, pairwise majority voting can produce a cycle: A defeats B, B defeats C, and C defeats A. The group then has no option that defeats every alternative, and agenda order can affect the final outcome.



HOW TO RECOGNIZE IT

- There are at least three alternatives and voters rank them differently.
- Compare alternatives two at a time using majority rule.
- The collective relation cycles even though each individual's ranking is consistent.
- Because no option is a universal majority winner, agenda order may matter.



WATCH OUT

A voting cycle does not require irrational voters. Individually transitive rankings can aggregate into an intransitive majority relation.



WORKED EXAMPLE: THREE VOTERS, THREE OPTIONS

DIAGNOSE

- pairwise majority
- consistent voters
- A beats B
- B beats C
- C beats A

Voter 1 ranks $A > B > C$; Voter 2 ranks $B > C > A$; Voter 3 ranks $C > A > B$. Against B, A wins with Voters 1 and 3. Against C, B wins with Voters 1 and 2. Against A, C wins with Voters 2 and 3. Thus the majority relation is $A > B > C > A$, so the agenda can determine which comparison occurs last and which option survives.



CHECK YOURSELF

Why can changing the order of pairwise votes change the winner when majority preferences form a cycle?

**READY?**

Return to the game and master this concept.